🚙 Accident Severity Analysis & Prediction

📌 Project Overview

This project focuses on analyzing road accident data to identify key factors affecting accident severity and building a simple predictive model to forecast severity levels.

The analysis includes **data cleaning, exploratory data analysis (EDA), hypothesis testing, and predictive modeling**, along with an interactive dashboard for better visualization.

🎯 Objectives

- Analyze accident data to find patterns and trends
- Identify factors influencing accident severity
- Perform hypothesis testing on key variables
- Build a predictive model for severity classification
- Create an interactive dashboard for insights

🧠 Key Questions

1. Does weather affect accident severity?
2. Does driver behavior impact accidents?
3. Does traffic influence severity?
4. What are the major patterns in accident data?
5. Can we predict accident severity?

🛠️ Tools & Technologies

- Python (Pandas, NumPy)
- Matplotlib / Seaborn
- Scikit-learn
- SQL
- Streamlit (Dashboard)
- Jupyter Notebook

📊 Dataset Summary

- Format: CSV
- Contains accident-related data
- Features include:
  - Weather condition
  - Driver behavior
  - Traffic level
  - Accident severity

🧹 Data Cleaning Steps

- Removed missing values
- Handled duplicates
- Converted data types
- Performed feature engineering

📈 Exploratory Data Analysis (EDA)

Instead of only static graphs, an **interactive dashboard** was created using Streamlit.

Dashboard Features:

- Filter data by location and conditions
- View accident trends
- Analyze severity distribution
- Real-time interaction

📊 Hypothesis Testing

Statistical testing was performed using the Chi-Square Test.

Results:

- Weather → Affects severity
- Driver Behavior → Impacts accidents

- Traffic → No strong relation

🤖 Predictive Analysis

A simple machine learning model was implemented:

- Model: Random Forest
- Input: Weather, Driver Behavior, Traffic
- Output: Predicted accident severity

📌 Key Insights

- Weather conditions influence accident severity
- Driver behavior plays an important role
- Traffic alone does not strongly impact severity
- Certain conditions lead to higher risk accidents

📊 Dashboard Preview

Interactive dashboard allows users to explore accident data dynamically and gain insights efficiently.

📃 Conclusion

This project successfully analyzed accident data and identified key influencing factors. A predictive model was also developed to estimate accident severity, making the analysis more practical and insightful.

⚠️ Limitations

- Limited dataset size
- Some missing or inconsistent data
- Model accuracy can be improved

🚀 Future Scope

- Use larger and real-time datasets
- Apply advanced machine learning models

- Deploy as a web application
- Integrate live traffic data

📚 References

- Dataset source (Kaggle or other)
- Python documentation
- Machine learning resources

🧑‍💻 Author

Your Name
(Data Analytics / Data Science Project)

⭐ If you like this project

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